Product Requirements Document (PRD)

StarGate - Face Recognition Attendance System

Version: 1.0
Product Name: StarGate
Platform: Desktop Application
Target Users: Sekolah, Universitas, Perusahaan, Pabrik, Instansi Pemerintah
Technology: Python Desktop Application (PyQt6)

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1. Executive Summary

Latar Belakang

Banyak institusi masih menggunakan absensi manual, fingerprint, RFID, atau QR Code yang memiliki beberapa kelemahan:

- Titip absen
- Antrean panjang
- Kontak fisik perangkat
- Sulit monitoring real-time
- Sulit melakukan audit kehadiran

StarGate hadir sebagai aplikasi absensi berbasis Face Recognition dan Anti Spoofing untuk memastikan identifikasi pengguna secara cepat, akurat, dan aman.

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2. Tujuan Produk

Tujuan Utama

Menyediakan sistem absensi modern menggunakan wajah yang:

- Cepat (< 2 detik)
- Akurat (> 98%)
- Anti Titip Absen
- Anti Foto
- Mudah digunakan
- Mendukung multi lokasi
- Dapat dipakai sekolah maupun perusahaan

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3. Target Market

Pendidikan

- TK
- SD
- SMP
- SMA
- Universitas
- Lembaga Kursus

Kebutuhan

- Absensi siswa
- Presensi guru
- Rekap keterlambatan
- Rekap kehadiran kelas

Perusahaan

- Kantor
- Pabrik
- Hotel
- Rumah Sakit
- Gudang
- Retail

Kebutuhan

- Attendance check-in/check-out
- Shift kerja
- Lembur
- Monitoring kehadiran

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4. Scope Product

In Scope

✅ Face registration

✅ Face recognition

✅ Liveness detection

✅ Attendance management

✅ Shift management

✅ Reporting

✅ Export Excel

✅ Export PDF

✅ User management

✅ Multi camera

✅ Multi group

✅ Audit log

Out of Scope (V1)

❌ Mobile App

❌ Cloud SaaS

❌ GPS Tracking

❌ Payroll

❌ Face Mask Recognition

❌ Visitor Management

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5. Functional Requirements

Module 1: Authentication

Login

Features

- Username login
- Password login
- Remember me
- Change password

User Roles

Role	Access
Super Admin	Full

Admin	Management
Operator	Attendance
Viewer	Report Only

Module 2: Dashboard

Dashboard Attendance

Widgets

Today Summary

- Total Registered Members
- Present
- Late
- Absent
- On Leave

Live Attendance Feed

Realtime activity:

```
08:10  Ahmad
08:11  Budi
08:12  Siti
```

Charts

- Attendance per Shift
- Attendance per Group
- Daily Trend
- Monthly Trend

Module 3: Register Member

Tujuan

Mendaftarkan wajah anggota baru.

Data Input

Field	Required
Member ID	Yes
Full Name	Yes
Group	Yes
Email	No
Phone	No
Position	No
Photo	Yes

Capture Face

Step

1. Open Camera
2. Detect Face
3. Capture Multiple Angles

Face Angles

- Front
- Left
- Right
- Up
- Down

Output

Generate:

```
Embedding Vector[512]
```

Disimpan ke PostgreSQL + pgvector.

Module 4: Members

Features

List Members

Filter:

- Name
- Group
- Active Status

Action

- Create
- Edit
- Delete
- View Attendance History
- Re-register Face

Module 5: Member Groups

Tujuan

Mengelompokkan anggota.

Contoh Sekolah

```
Kelas 10A
Kelas 10B
Kelas 11A
```

Contoh Perusahaan

```
HR
IT
Finance
Production
```

Features

- Create Group
- Edit Group
- Delete Group

Module 6: Devices

Device Management

Supported

- Internal Webcam
- USB Webcam
- IP Camera (RTSP)

Features

Add Device

```
Device Name
Camera Type
RTSP URL
Resolution
FPS
```

Test Connection

Preview stream sebelum digunakan.

Module 7: Attendance

Overview

Modul inti StarGate.

Workflow

```
Camera
   ↓
Face Detection
   ↓
Liveness Check
   ↓
Face Recognition
   ↓
Attendance Validation
    ↓
 Store Attendance
```

Attendance Types

Check In

```
IN
```

Check Out

```
OUT
```

Break Start

```
BREAK_OUT
```

Break Finish

```
BREAK_IN
```

Rules

Cooldown

Default:

```
60 detik
```

Untuk mencegah spam attendance.

Duplicate Prevention

Jika user sudah check-in:

```
Reject duplicate
```

Feedback

Success

```
✅ Welcome Ahmad
Check In Success
```

Failed

```
❌ Face Not Recognized
```

```
❌ Spoof Detected
```

Module 8: Shift Management

Features

Create Shift

Contoh:

Shift	Start	End
Pagi	08:00	17:00
Malam	20:00	04:00

Rules

- Grace Period
- Late Calculation
- Early Leave

Module 9: Reports

Attendance Reports

Filters

- Date From
- Date To
- Group
- Member
- Shift
- Attendance Status

Report Types

Daily Report

Weekly Report

Monthly Report

Custom Report

Export

Excel

Format:

```
1
```

Menggunakan:

```
1  Pandas
2  OpenPyXL
```

PDF

Format:

```
1
```

Menggunakan:

```
1  ReportLab
```

Module 10: User Management

Features

User CRUD

Field	Required
Username	Yes
Full Name	Yes
Role	Yes
Password	Yes

Permission Matrix

Feature	Super Admin	Admin	Operator	Viewer
Members	✅	✅	✅	❌
Attendance	✅	✅	✅	❌
Reports	✅	✅	✅	✅
Settings	✅	✅	❌	❌
Users	✅	❌	❌	❌

Module 11: Settings

System Settings

Recognition

Parameter	Default
Similarity Threshold	0.60
Max Face Size	Auto
Recognition Timeout	3 sec

Anti Spoof

Parameter	Default
Enable	Yes
Threshold	0.85

Attendance

Parameter	Default
Duplicate Window	60s
Check-In Start	Shift
Auto Checkout	Optional

Non Functional Requirements

Performance

Metric	Target
Face Detection	< 300 ms
Recognition	< 500 ms
Attendance Save	< 200 ms
Startup App	< 5 sec

Reliability

Availability:

```
99.5%
```

Security Requirements

Authentication

- BCrypt Password Hashing
- Session Timeout
- Account Lockout

Database

- Parameterized Query
- SQLAlchemy ORM
- No raw SQL

Audit Log

Log seluruh aktivitas:

- Login
- Attendance
- User Create
- User Delete
- Settings Change

Database Design

members

```
id
member_code
full_name
group_id
email
phone
position
photo_path
is_active
 created_at
 updated_at
```

member_groups

```
id
group_name
description
created_at
```

face_embeddings

```
id
member_id
embedding vector(512)
created_at
```

shifts

```
id
shift_name
start_time
end_time
grace_minutes
```

attendance

```
id
member_id
device_id
shift_id
```

```
attendance_type
attendance_time
status
```

devices

```
id
device_name
device_type
rtsp_url
resolution
fps
```

users

```
id
username
password_hash
fullname
role
last_login
```

settings

```
id
setting_key
setting_value
```

Technical Architecture

Application Layer

```
PyQt6
├── Dashboard
├── Attendance
├── Reports
├── Members
├── Settings
└── Users
```

Recognition Layer

```
RetinaFace
```

```
├── Face Detection
│
InsightFace ArcFace
├── Embedding
│
pgvector Search
├── Similarity Match
│
 MiniFASNet
 └── Liveness Detection
```

Data Layer

```
SQLAlchemy ORM
        ↓
PostgreSQL
        ↓
pgvector
```

Folder Structure

```
stargate/
│
├── app/
│   ├── ui/
│   ├── pages/
│   ├── widgets/
│   ├── services/
│   ├── repositories/
│   ├── models/
 │   ├── reports/
 │   ├── exports/
 │   └── settings/
 │
├── ai/
 │   ├── detector/
 │   ├── recognizer/
 │   ├── liveness/
 │   └── embedding/
 │
├── database/
 │   ├── migrations/
 │   ├── schema/
```

```
23  │   └── seed/
24  │
25  ├── logs/
26  │
27  ├── resources/
28  │
29  └── main.py
```

MVP Success Metrics

Recognition

- Accuracy ≥ 98%

Liveness

- Spoof Detection ≥ 95%

Attendance Speed

- ≤ 2 detik

Database Search

- ≤ 100ms untuk 10.000 member

Future Roadmap (V2)

Mobile Companion App

- Android
- iOS

WhatsApp Notification

- Check-In Notification
- Absent Notification

Payroll Integration

- Overtime
- Salary Calculation

Cloud Synchronization

- Multi Branch
- Centralized Dashboard

API Gateway

- REST API
- GraphQL API

Kesimpulan

StarGate adalah aplikasi desktop absensi berbasis AI Face Recognition yang menggabungkan **RetinaFace + ArcFace + MiniFASNet + PostgreSQL pgvector** untuk menghasilkan sistem absensi yang cepat, akurat, anti-spoofing, dan siap digunakan baik untuk sekolah maupun perusahaan. Arsitektur dirancang modular sehingga mudah dikembangkan menjadi solusi multi-cabang, cloud, maupun terintegrasi dengan payroll dan HRIS di masa depan.